

Groups (section 4)

How do we solve $5+x=2$ for $x \in \mathbb{Z}$?

$$5+x=2$$

$$-5+(5+x)=-5+2$$

$$(-5+5)+x=-3$$

$$0+x=-3$$

$$x=-3$$

associativity

inverses

identity

Def: A **group** is a binary algebraic structure $\langle G, * \rangle$ such that the following axioms are satisfied:

(1) $*$ is associative:

$$\forall a, b, c \in G \text{ we have } a * (b * c) = (a * b) * c.$$

(2) there is an identity element $e \in G$:

$$\forall a \in G \text{ we have } e * a = a * e = a$$

(3) each $a \in G$ has an **inverse** $a' \in G$: OR a^{-1}

$$a * a' = a' * a = e.$$

Example: $\langle G, * \rangle = \langle \mathbb{Z}_3, +_3 \rangle$

$$a' * a = e$$

$$a' +_3 a = 0$$

$+_3$	0	1	2
0	0	1	2
1	1	2	0
2	2	0	1

Identity: 0 is the identity

Inverse: $0' = 0$

$$1' = 2 \quad \text{since } 2 + 1 = 0$$

$$2' = 1 \quad \text{since } 1 + 2 = 0$$

Prop In a group G ,
we have $e' = e$.

Pf: $e * e = e$

Associativity: follows from associativity
for $\langle \mathbb{Z}, + \rangle$.

Abelian groups

Def: A group G is called **abelian** if $*$ is commutative:

$$\forall a, b \in G \text{ we have } a * b = b * a.$$

More (non)-examples

Which are groups? Abelian groups?

$\langle \mathbb{Z}^+, + \rangle$:
associative ✓
identity ✗
~~inverses~~

is there $e \in \mathbb{Z}^+$ with $e+b=b, \forall b$.

$\langle \mathbb{Z}_{\geq 0}, + \rangle$:
associative ✓
identity ✓
inverses ✗

0 is the identity
no negative numbers

$\langle \mathbb{Q}^+, \cdot \rangle$:
associative ✓
identity ✓
inverses ✓

1 is the identity
the inverse of $\frac{a}{b}$ is $\frac{b}{a}$ since $\frac{a}{b} \cdot \frac{b}{a} = 1$

$\langle \mathbb{Q}, \cdot \rangle$:
no inverse
for 0

abelian ✓

More (non)-examples:

$$\langle \mathbb{Z}^+, \cdot \rangle:$$

associative ✓

identity ✓

inverses ✗

1 is an identity

2 has no inverse since $\frac{1}{2} \notin \mathbb{Z}$

$$\langle GL_2(\mathbb{C}), \cdot \rangle := \{ 2 \times 2 \text{ invertible matrices with entries from } \mathbb{C} \}$$

↑ "general linear" group

associative ✓

• matrix mult. is assoc.

identity ✓

$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

• functional comp. is assoc.

inverses ✓

matrix inverse

abelian ✗

Trivial group:

*	e
e	e

Thm (Cancellation law for groups)

Let $\langle G, * \rangle$ be a group and $a, b, c \in G$. Then

$$a * b = a * c \Rightarrow b = c.$$

Pf: $\S$ $a * b = a * c$

← inverses

$$a' * (a * b) = a' * (a * c)$$

← associativity

$$(a' * a) * b = (a' * a) * c$$

$$e * b = e * c$$

← identity

$$b = c$$



Exercise:

State and prove

"right" sided
cancellation law.

Thm let $\langle G, * \rangle$ be a group and $a, b \in G$. Then $a * x = b$ has a unique solution x in G .

Pf:

Existence: $x = a' * b$ is a solution, since

$$a * (a' * b) \stackrel{\text{assoc.}}{=} (a * a') * b = e * b \stackrel{\text{identity}}{=} b.$$

Uniqueness: $\nexists$ $a * y = b$ then $a * x = a * y$

Cancellation Law

$\Rightarrow$

$$x = y.$$



Thm An element a in a group $\langle G, * \rangle$ has a unique inverse.

Pf: Any inverse is a solution to $a * x = e$,
by the previous Thm, this has a unique solution.

Cor: For $a, b \in G$, we have $(a * b)' = b' * a'$.

Pf: Need to check $(a * b) * (b' * a')$ is e .

$$\begin{aligned} (a * (b * b')) * a' &= (a * e) * a' \\ &= a * a' = e. \end{aligned}$$

Exercise:
Show that
a right inverse
of a is also
a left
inverse.

Group tables

What can the table of a group look like?

$*$	e	x_1	x	x_k
e	e	x_1	$\dots$	x_k
x_1	x_1			
a	$\vdots$		b	
x_k	x_k			

Identity: first row & column

Unique solution to $a * x = b$: each element $g \in G$ appears exactly once in each row and in each column.

Associativity: ?

(Abelian): symmetric across main diagonal